Van Dai Do

APPLIED ARTIFICIAL INTELLIGENCE INITIATIVE (A2I2), DEAKIN UNIVERSITY

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Education

Applied Artificial Intelligence Initiative, Deakin University, PhD. in Computer Science — Aug 2023 – Jan 2026

- Research Topic: *Efficient and Safe Large Language Models With Reinforcement Learning*

Hanoi University of Science and Technology, BS in Electronics and Telecommunications - class for the gifted — Sep 2018 – Aug 2022

- GPA: 3.4/4.0
- Thesis: *Optimizing Edge Computing Models Using Deep Reinforcement Learning*

Hanoi-Amsterdam High School For The Gifted, Major: Mathematics — Sep 2015 – Aug 2018

- GPA: 9.0/10

Research & Professional Experience

Associate Research Fellow, Deakin University – Waurn Ponds, Australia — Jan 2026 – Present

- Research on reasoning capabilities of Memory and Language Models.
- Research on and develop foundational time series models.
- Proposed a retrieval-based latent steering method that improves fine-tuned decoder-only time-series forecasting models at inference time. **NeurIPS 2026**

Research Assistant, Deakin University – Waurn Ponds, Australia — Feb 2025 – Jan 2026

- Research on time series forecasting under Dr. Hung Le's DECRA grant funded by Australian Research Council: *Complex Time-series System Forecasting Reinforced by Expert Knowledge*.
- Developed a universal error-correction framework for deep time-series forecasting, mitigating autoregressive error accumulation via a lightweight seasonal-trend decomposition corrector that improves long-horizon forecasts. **ICML 2026**
- Developed a physics-informed time-series forecasting framework for molecular dynamics simulations, predicting atomic trajectories via displacements and enforcing physical plausibility through DFT-based potential functions. **ICDM 2025**
- Research on reasoning capabilities of Language Models by combining intrinsic reward with Reinforcement Learning finetuning of Language Models. **TMLR**

Ph.D Research on Efficient and Safe Large Language Models, Deakin University – Waurn Ponds, Australia — Aug 2023 – Jan 2026

- Delivered a tutorial on improving Reinforcement Learning (RL) exploration efficiency through memory-driven intrinsic motivation, presenting a taxonomy of curiosity-based agents and exploring recent advances using language models and causal reasoning for exploration. **Tutorial - AAMAS 2024**
- Developed a RL–based prompt optimization approach that efficiently selects few-shot examples from episodic memory to enhance language model performance and generalization across diverse NLP downstream tasks. **ECAI 2024 (Oral)**
- Developed a training-free framework for token-level activation steering in LLMs, using episodic memory and nearest-neighbor retrieval to provide adaptive alignment across safety, style transfer, and role-playing tasks. **ACL 2025**
- Developed a non-parametric alignment framework that leverages episodic control to guide LLM outputs during inference, achieving effective alignment with minimal human feedback or fine-tuning. **EMNLP 2025**
- Developed RL-based method for selecting training examples for efficient training of LLMs. **ACL 2026**

AI Research Engineer, VinBigdata – Hanoi, Vietnam Sep 2022 – Aug 2023

- Designed and implemented passage retrieval and aspect-based sentiment extraction systems for both Vietnamese and English languages.
- Implemented Deep Passage Retriever for Open-Domain QA System for Vietnamese questions.

AI Research Intern, CinnamonAI – Hanoi, Vietnam Jun 2022 – Aug 2022

- Agile software development (SCRUM, Git, SOLID, Docker)
- Designed and developed the text recognition module in a table information extraction.

Grants

Lambda's Research Grant Program 2026

Selected Honours And Awards

Kingston AI Group's Symposium Poster Presentation - Kingston AI Group 2026
Poster presentation (ACL 2025 poster): "Dynamic Steering With Episodic Control".

Deakin Applied Artificial Intelligence Initiative - 3MT Heat 2025
Winner of 3 Minute Thesis Contest, Deakin Applied Artificial Intelligence Initiative 2025. Title: Word-by-word: Better responses with memory.

Deakin University Postgraduate Research Scholarship 2023
Offered Higher Degree by Research scholarship for bachelor students

Honors Distinction 2022
Thesis: Optimizing Edge Computing Models Using Deep Reinforcement Learning

CinnamonAI - Minimum Viable Product (MVP) Competition 2022
2nd place in Minimum Viable Product competition with Table Information Extraction system.

WorldQuant - WorldQuantChallenge 2019 2019
Bronze medalist (solo) in WorldQuantChallenge 2019

Hanoi Mathematics Olympiad for Students 2018
The annual Vietnamese regional mathematics competition for high school students

- Third prize

Selected Publications (Total Citations: 93, H-Index: 5)

1. ***Van Dai Do***, Huu Hiep Nguyen, Minh Hoang Nguyen, Hung Le, "SteerCast: Retrieval-Based Latent Steering for Decoder-Only Time Series Forecasting". **NeurIPS, 2026**
2. Minh Hoang Nguyen, ***Van Dai Do***, Huu Hiep Nguyen, Dung Nguyen, Kien Do, Hung Le, "Reviving Error Correction in Modern Deep Time-Series Forecasting" - **ICML 2026**
3. ***Van Dai Do***, Manh Nguyen, Svetha Venkatesh, Hung Le, "SPaCe: Unlocking Sample-Efficient Large Language Models Training With Self-Pace Curriculum Learning" - **ACL 2026**
4. Hung Le, ***Van Dai Do***, Dung Nguyen, Svetha Venkatesh, "Reasoning Under 1 Billion: Memory-Augmented Reinforcement Learning for Large Language Models" - **TMLR 2025**
5. Hung Le, Sherif Abbas, Minh Hoang Nguyen, ***Van Dai Do***, Huu Hiep Nguyen, Dung Nguyen, "Accelerating Long-Term Molecular Dynamics with Physics-Informed Time-Series Forecasting" - **ICDM 2025**
6. ***Van Dai Do***, Quan Tran, Ahmed Kirmani, Lu Zhang, Hung Le, "Sample Efficient Alignment Learning With Episodic Control" - **EMNLP 2025**
7. ***Van Dai Do***, Quan Tran, Svetha Venkatesh, Hung Le, "Dynamic Steering With Episodic Memory For Large Language Models" - **ACL 2025 [pdf] [code]**
8. ***Van Dai Do***, Quan Tran, Svetha Venkatesh, Hung Le, "Large Language Models Prompting With Episodic Memory" - **ECAI 2024 (Oral) [pdf] [code]**
9. Hung Le, Hoang Nguyen, ***Van Dai Do***, "Unlocking Exploration: Self-Motivated Agents Thrive on Memory-Driven Curiosity" - **AAMAS Tutorials, 2024**

Invited Talks

- "Advances in Multi-Modal Time Series" - ACML 2026
- "Dynamic Steering With Episodic Control" - Kingston AI Symposium 2026
- "Unlocking Exploration: Self-Motivated Agents Thrive on Memory-Driven Curiosity" - AAMAS 2024

Skills

- Strong foundation in research methodology, with the ability to design, analyze, and interpret both theoretical and empirical studies.
- Broad experience across AI/ML and Large Language Models, Time Series research, encompassing data analysis, model development, evaluation, and optimization, with a keen interest in advancing state-of-the-art performance on public benchmarks.
- Capable of conducting independent research as well as leading and collaborating effectively within research teams.

Languages

Vietnamese
Fluent (native speaker)

French 2022
DELF (diplôme d'études en langue française) - A2

English 2021
IELTS - Overall band: 8.0

Referees

Professor Svetha Venkatesh
Deakin University
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Dr. Hung Le
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